Architecture and Workflow

MQTT follows a client-broker architecture, where communication occurs via a central server called the broker. Clients can act as publishers, sending messages to specific topics, or subscribers, receiving messages from topics they are interested in. The broker manages all communication, ensuring messages from publishers are delivered to the appropriate subscribers.

The typical workflow involves:

1. **Client Connection:** The client establishes a connection with the broker using a CONNECT packet, optionally providing authentication credentials.
2. **Subscription:** Clients subscribe to topics of interest using a SUBSCRIBE packet.
3. **Publishing:** Publishers send messages to topics via the PUBLISH packet.
4. **Message Delivery:** The broker forwards the messages to all subscribers of the topic, according to the QoS level.
5. **Disconnection:** Clients can disconnect gracefully using the DISCONNECT packet, while the broker handles unexpected disconnections using the Last Will and Testament feature.

This architecture allows MQTT to support asynchronous communication, ensuring that publishers and subscribers do not need to be online simultaneously.